DIFFERENTIAL EQUATIONS - MATH 251

HIGHER ORDER ODE

M. A. BOATENG (Ph.D., MIMA)



KWAME NKRUMAH UNIVERSITY
OF SCIENCE AND TECHNOLOGY

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HIGHER ORDER ODEs

1 INTRODUCTION

An nth order ODE is of the form:

$$a_n(x)y^{(n)} + a_{(n-1)}(x)y^{(n-1)} + \ldots + a_2(x)y^{(2)} + a_1(x)y' + a_0(x)y = F(x)$$

If $F(x) = 0$, the equation is homogeneous, if $F(x) \neq 0$, the equation is non-homogeneous.

The solution of the homogeneous DE is called the Complimentary Function, y_c and the solution of the non-homogeneous part is called the Particular Integral, y_p.

The General Solution is thus:

$$y = y_c + y_p.$$



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Homogeneous Equations with Constant Coefficients.

2 HOMOGENEOUS EQUATIONS WITH CONSTANT COEFFICIENTS

Let $ay'' + by' + cy = 0$ be a second order homogeneous ODE with constant coefficients, i.e. a, b and c are constants.

let $y(x) = e^{rx}$ be a trial solution, then ; $y'(x) = re^{rx}$; $y''(x) = r^2e^{rx}$
$ar^2e^{rx} + bre^{rx} + ce^{rx} = 0$
$e^{rx}(ar^2 + br + c) = 0$
$e^{rx} \neq 0$,
$ar^2 + br + c = 0$ which is a quadratic equation (called an auxilliary or characteristic equation).

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **CASE 1**: $b^2 - 4ac > 0$, Real but distinct roots.
  $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$

- **CASE 2**: $b^2 - 4ac = 0$, Real and equal roots.
  $y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$

- **CASE 3**: $b^2 - 4ac < 0$,Complex roots.
  $y(x) = e^{\alpha x}(c_1 cos\beta x + c_2 sin\beta x)$
  $\alpha = \frac{-b}{2a}, \beta = \frac{\sqrt{b^2-4ac}}{2a}$



EXAMPLES

2 HOMOGENEOUS EQUATIONS WITH CONSTANT COEFFICIENTS

1. Solve $y'' + 2y' + y = 0$

SOLUTION

let $y(x) = e^{rx}$

$y'(x) = re^{rx}$

$y''(x) = r^2 e^{rx}$

$r^2 e^{rx} + 2re^{rx} + e^{rx} = 0$

$e^{rx}(r^2 + 2r + 1) = 0$

$r^2 + 2r + 1 = 0$

$r_1 = r_2 = -1$

$\implies y(x) = c_1 e^{-x} + c_2 x e^{-x}$

2. $y'' + 2y' + 5y = 0$

SOLUTION

let $y(x) = e^{rx}$

$y'(x) = re^{rx}$

$y''(x) = r^2e^{rx}$

$r^2e^{rx} + 2re^{rx} + 5e^{rx} = 0$
v $e^{rx}(r^2 + 2r + 5) = 0$

$r^2 + 2r + 5 = 0$
v $r = -1 \pm 2i$

$\implies y(x) = e^{-x}(c_1 cos2x + c_2 sin2x)$

3. $y'' + y' - 6y = 0$

$r^2 + r - 6 = 0$

$y(x) = c_1 e^{2x} + c_2 e^{-3x}$

EXERCISE

Solve the following differential equations

- $2y'' + 4y' + 4y = 0$
- $y'' - 6y' + 9y = 0$
- $y'' - y' - 6y = 0$



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Consider the non-constant coefficient, second order ODE of the form:

$$y'' + p(x)y' + q(x)y = 0,$$

Principle of Superposition

Let y_1 and y_2 be two solutions of the homogeneous linear equation above on the interval I. If c_1 and c_2 are constants, then the linear combination: $y = c_1y_1 + c_2y_2$ is also a solution to the DE on I.



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Linear Independence of Two Functions and the Wronskian

Two functions defined on an open interval are called linearly independent provided that neither is a constant multiple of the other.

Two functions are said to be linearly dependent if they are not linearly independent, i.e one is a constant multiple of the other.

In general, a set of functions $[y_1(x), y_2(x), \ldots, y_n(x)]$ is linearly dependent on $a \leq x \leq b$ if there exist constants $c_1, c_2, \ldots, c_n$, not all zero, such that:
$c_1y_1 + c_2y_2 + \ldots + c_ny_n \equiv 0$ on $a \leq x \leq b$.



Example

$y_1 = cosx$ and $y_2 = sinx$ are two solutions of the equation $y'' + y = 0$. By the principle of superposition: $y = 3y_1 - 2y_2 = 3cosx - 2sinx$ is also a solution.

Given two functions f and g, the Wronskian is the determinant:

$$W(f,g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix}$$

$= fg' - f'g$



Example:

$$W(cosx, sinx) = \begin{vmatrix} cosx & sinx \\ -sinx & cosx \end{vmatrix}$$

$$= cos^2x + sin^2x = 1 \neq 0$$

Given two functions $f(x)$ and $g(x)$ that are differentiable on some interval I:

- If $W(f(x), g(x)) \neq 0$ for some x_0 on I, then $f(x)$ and $g(x)$ are linearly independent on the interval.
- If $f(x)$ and $g(x)$ are linearly dependent on I, then $W(f(x), g(x)) = 0$ for all x in the interval.

Determine if the following sets of functions are linearly dependent or linearly independent:

- $f(x) = 9cos2x, g(x) = 2cos^2x - 2sinx$
- $f(x) = 2x^2, g(x) = x^4$



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Reduction of Order

5 REDUCTION OF ORDER

Consider the non-constant coefficient, second order ODE of the form:
$y'' + p(x)y' + q(x)y = 0$

- To use the method of reduction of order, we must be given one solution to the DE.
- We start by assuming the second solution will be of the form: $y_2(x) = u(x).y_1(x)$

The general formula to find y_2 is given as:

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int P(x)dx} dx$$

refer to proof in the lecture notes

EXAMPLES

Solve the following:

- $x^2y'' - 3xy' + 4y = 0$ given $y_1 = x^2$.
- $2x^2y'' + xy' - 3y = 0$ given $y_1(x) = x^{-1}$.

SOLUTION

Break into groups and solve.

TRIAL

Solve the following differential equations

1. $(x^2 - 1)y'' - 2xy' + 2y = 0, y_1 = x$
2. $x^2y'' - 3xy' + 4y = 0, y_1 = x^2$



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METHOD OF UNDETERMINED COEFFICIENT

6 METHOD OF UNDETERMINED COEFFICIENT

Given $ay'' + by' + cy = f(x)$

$y = y_c(x) + y_p(x)$

where y_c =Complimentary Solution, y_p = Particular Integral

1. **CASE 1:** $f(x) = ae^{bx}$
   $y_p = Ax^k e^{bx}, k = 0$ if $r \neq b$

2. **CASE 2:** $f(x) = asinbx$ or $acosbx$
   $y_p = x^k(Asinbx + Bcosbx), k = 0$ if $r^2 + b^2$ is **NOT** a factor.

3. **CASE 3:** $f(x) = ax^n$
   $y_p = x^k(Ax^n + Bx^{n-1} + \ldots + Px + Q)$
   $k = 0$ if $r = 0$ is not a root.



Examples

6 METHOD OF UNDETERMINED COEFFICIENT

1. solve $y'' - 3y' + 2y = e^{2x}$

SOLUTION

$r^2 - 3r + 2, r_1 = 1, r_2 = 2$

$y_c = c_1 e^x + c_2 e^{2x}$

let $y_p = Axe^{2x}$ since $f(x) = e^{2x}$

$y' = Ae^{2x} + 2Axe^{2x}$

$y'' = 2Ae^{2x} + 4Axe^{2x} + 2Ae^{2x}$

$y'' = 4Ae^{2x} + 4Axe^{2x}$

$4Ae^{2x} + 4Axe^{2x} - 3(Ae^{2x} + 2Axe^{2x}) + 2Axe^{2x} = e^{2x}$

$Ae^{2x} = e^{2x}$

$A = 1$

$y(x) = c_1 e^x + c_2 e^{2x} + xe^{2x}$

2. Solve $y'' + 4y = 2\sin 2x$

SOLUTION

$r^2 + 4 = 0, r = \pm 2i$

$y_p = Axsin2x + Bxcos2x$

$y' = Asin2x + 2Axcos2x + Bcos2x - 2Bxsin2x$

$y'' = 2Acos2x + 2Acos2x - 4Asin2x - 2Bsin2x - 2Bsin2x - 4Bxcos2x$

substituting into the equation yields;

$2Acos2x + 2Acos2x - 4Axsin2x - 2Bsin2x - 2Bsin2x - 4Bxcos2x + 4(Axsin2x + Bxcos2x) = 2\sin 2x$

$A = 0, B = \frac{-1}{2}$

$y_p = -\frac{-1}{2}xsin2x$

$y(x) = -\frac{-1}{2}xsin2x + c_1 sin2x + c_2 cos2x$

3. Solve $y'' + 4y = 3x^3$

SOLUTION

$r^2 + 4 = 0, r = \pm 2i$

$y_p = Ax^3 + Bx^2 + Cx + D$

$y' = 3Ax^2 + 2Bx + C$

$y'' = 6Ax + 2B$

$6Ax + 2B + 4(Ax^3 + Bx^2 + Cx + D) = 3x^3$

$4Ax^3 + 4Bx^2 + (6A + 4C)x + 2B + 2D = 3x^3$

Comparing term by term;

$A = \frac{3}{4}, B = 0, C = -\frac{9}{8}$

$y(x) = c_1 sin2x + c_2 cos2x + \frac{3}{4}x^3 - \frac{9}{8}x$

TRIAL

Solve the following undetermined coefficient differential equations

- $y'' + 3y' + 2y = 2x^2 + 4x + 5$
- $y'' - 2y' - 3y = 3e^{2t}$
- $y'' - 3y' + 2y = e^t sin(t)$



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METHOD OF VARIATION OF PARAMETERS

7 METHOD OF VARIATION OF PARAMETERS

Consider the differential equation, $y'' + q(t)y' + r(t)y = g(t)$. Assuming that $y_1(t)$ and $y_2(t)$ are fundamental solutions to $y'' + q(t)y' + r(t)y = 0$
Then the particular solution to the non-homogeneous differential equation is,

$$y_p = -y_1 \int \frac{y_2 g(t)}{W(y_1, y_2)} dt + y_2 \int \frac{y_1 g(t)}{W(y_1, y_2)} dt$$

Where

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1^l & y_2^l \end{vmatrix}$$

is called the Wronskian.

EXAMPLE

Find the general solution to $ty'' - (t+1)y' + y = t^2$
Given $y_1(t) = e^t$ and $y_2(t) = t + 1$

SOLUTION

First we divide through by t

$y'' - (1 + \frac{1}{t})y' + \frac{1}{t}y = t$

The Wronskian for the fundamental set of solutions is

$$W = \begin{vmatrix} e^t & t+1 \\ e^t & 1 \end{vmatrix}$$

$=e^t - e^t(t+1)$

$= -te^t$

The particular solution becomes

$$y_p = -e^t \int \frac{(t+1)t}{-te^t} dt + (t+1) \int \frac{e^t t}{-te^t} dt$$

$$= e^t \int (t+1)e^{-t} dt - (t+1) \int dt$$

$$= e^t(-e^{-t}(t+2)) - (t+1)t$$

$$= -t^2 - 2t - 2$$

The general solution then becomes

$$y(t) = C_1 e^t + C_2(t+1) - t^2 - 2t - 2$$



ASSIGNMENT

7 METHOD OF VARIATION OF PARAMETERS

Solve the following second order DE using the method of variation of parameters:

- $y'' - 2y' + y = \frac{e^t}{t^2+1}$
- $2y'' - y' - y = 2e^t$



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Thank you for listening!
Any questions?


thank you!